

Chapter 11

Iterated Dominance and Nash Equilibrium

In the previous chapter we examined simultaneous move games in which each player had a dominant strategy; the Prisoner's Dilemma game was one example. In many games, however, one or more players do not have dominant strategies. This chapter explores two solution concepts that we can use to analyze such games.

The first solution concept, iterated dominance, is a **refinement** of the dominant strategies approach from the previous chapter, meaning that iterated dominance is a stronger technique that builds upon (or refines) the results of the dominant strategies approach. In other words: the idea of dominant strategies often allows us to narrow down our prediction for the outcome of a game; iterated dominance allows us to narrow down our prediction at least as far, and sometimes further.

Unfortunately, this extra strength does not come for free. While dominant strategies is a reasonably simple idea, iterated dominance is (while not exactly a Nobel-prize-winning concept) one step closer to rocket science. As such, it requires more powerful assumptions about the intellectual capabilities of the optimizing individuals who are playing the games.

The second solution concept in this chapter, Nash equilibrium, is a refinement of iterated dominance: Nash equilibrium allows us to narrow down our prediction at least as far as iterated dominance, and sometimes further. Again, this extra strength does not come for free. Nonetheless, Nash equilibrium is one of the central concepts in the study of strategic behavior—a fact which helps explain why Nash equilibrium *is* a Nobel-prize-winning concept.

11.1 Iterated Dominance

The transition from dominant strategies to iterated dominance involves two ideas. The first is this: even when a player doesn't have a dominant strategy

(i.e., a *best* strategy, regardless of what the other players do), that player might still have one strategy that dominates another (i.e., a strategy A that is *better* than strategy B, regardless of what the other players do). As suggested by the terms “best” and “better”, the difference here is between a superlative statement (e.g., “Jane is the best athlete in the class”) and a comparative statement (“Jane is a better athlete than Ted”); because comparatives are weaker statements, we can use them in situations where we might not be able to use superlatives.

For example, consider the game in Figure 11.1. First note that there are no strictly dominant strategies in this game: U is not the best strategy for Player 1 if Player 2 plays L or C, M is not the best strategy for Player 1 if Player 2 plays R, and D is not the best strategy for Player 1 if Player 2 plays L or C. Similarly, L is not the best strategy for Player 2 if Player 1 plays U or D, C is not the best strategy for Player 2 if Player 1 plays M, and R is not the best strategy for Player 2 if Player 1 plays U, M, or D.

Although there are no strictly dominant strategies, we can see that no matter what Player 1 does, Player 2 always gets a higher payoff from playing L than from playing R. We can therefore say that L **strictly dominates** R for Player 2, or that R is **strictly dominated** by L for Player 2. (Note that we cannot say that L is a **strictly dominant** strategy for Player 2—it does not dominate C—but we can say that R is a **strictly dominated** strategy for Player 2: an optimizing Player 2 would *never* play R.)

The second idea in the transition from dominant strategies to iterated dominance is similar to the backward induction idea of anticipating your opponents’ moves: players should recognize that other players have strictly dominated strategies, and should act accordingly. In our example, Player 1 should recognize that R is a strictly dominated strategy for Player 2, and therefore that there is no chance that Player 2 will play R. In effect, the game now looks like that shown in Figure 11.2 on the next page: the lines through the payoffs in the R column indicate that both players know that these payoffs have no chance of occurring because R is not a viable strategy for Player 2.

But now we see that Player 1 has an obvious strategy: given that Player 2 is never going to play R, Player 1 should always play M. Once R is out of the way, U and D are both dominated by M for Player 1: regardless of whether Player 2 plays L or C, Player 1 always gets his highest payoff by playing M. This is the

		Player 2		
		L	C	R
Player 1	U	1, 10	3, 20	40, 0
	M	10, 20	50, -10	6, 0
	D	2, 20	4, 40	10, 0

Figure 11.1: A game without dominant strategies

		Player 2		
		L	C	R
Player 1	U	1, 10	3, 20	40, 0
	M	10, 20	50, -10	6, 0
	D	2, 20	4, 40	10, 0

Figure 11.2: Eliminating R, which is strictly dominated by L for Player 2

idea of **iteration**, i.e., repetition. Combining this with the idea of dominated strategies gives us the process of **iterated dominance**: starting with the game in Figure 11.1, we look for a strictly dominated strategy; having found one (R), we eliminate it, giving us the game in Figure 11.2. We then repeat the process, looking for a strictly dominated strategy in that game; having found one (or, actually two: U and D), we eliminate them. A final iteration would yield (M, L) as a prediction for this game: knowing that Player 1 will always play M, Player 2 should always play L.

A complete example

Consider the game in Figure 11.3 below. There are no strictly dominant strategies, but there is a strictly dominated strategy: playing U is strictly dominated by D for Player 1. We can conclude that Player 1 will never play U, and so our game reduces to the matrix in Figure 11.4a on the next page.

But Player 2 should know that Player 1 will never play U, and if Player 1 never plays U then some of Player 2's strategies are strictly dominated! Namely, playing L and playing R are both strictly dominated by playing C as long as Player 1 never plays U. So we can eliminate those strategies for Player 2, yielding the matrix in Figure 11.4b. Finally, Player 1 should anticipate that Player 2 (anticipating that Player 1 will never play U) will never play L or R, and so Player 1 should conclude that M is strictly dominated by D (the matrix in Figure 11.4c). Using iterated strict dominance, then, we can predict that Player 1 will choose D and Player 2 will choose C.

		Player 2		
		L	C	R
Player 1	U	1,1	2,0	2,2
	M	0,3	1,5	4,4
	D	2,4	3,6	3,0

Figure 11.3: Iterated strict dominance example

	L	C	R		L	C	R		L	C	R
U	1/1	2/0	2/2	U	1/1	2/0	2/2	U	1/1	2/0	2/2
M	0,3	1,5	4,4	M	0/3	1,5	4/4	M	0/3	1/5	4/4
D	2,4	3,6	3,0	D	2/4	3,6	3/0	D	2/4	3,6	3/0
(a)				(b)				(c)			

Figure 11.4: Solution to iterated strict dominance example

Question: Does the order of elimination matter?

Answer: Although it is not obvious, the end result of iterated strict dominance is always the same regardless of the sequence of eliminations. In other words, if in some game you can either eliminate U for Player 1 or L for Player 2, you don't need to worry about which one to "do first": either way you'll end up at the same answer.

A side note here is that this result only holds under iterated **strict** dominance, according to which we eliminate a strategy only if there is some other strategy that yields payoffs that are *strictly higher* no matter what the other players do. If you eliminate a strategy when there is some other strategy that yields payoffs that are *higher or equal* no matter what the other players do, you are doing iterated **weak** dominance, and in this case you will *not* always get the same answer regardless of the sequence of eliminations. (For an example see problem 10.) This is a serious problem, and helps explain why we focus on iterated strict dominance.

11.2 Nash Equilibrium

Tenuous as it may seem, iterated strict dominance is not a very strong solution concept, meaning that it does not yield predictions in many games. An example is the game in Figure 11.5: there are no strictly dominant strategies and no strictly dominated strategies.

So game theorists have come up with other solution concepts. The most important one is called **Nash equilibrium** (abbreviated NE). A Nash equi-

		Player 2		
		L	C	R
Player 1	U	5,1	2,0	2,2
	M	0,4	1,5	4,5
	D	2,4	3,6	1,0

Figure 11.5: Nash equilibrium example

librium occurs when the strategies of the various players are best responses to each other. Equivalently but in other words: given the strategies of the other players, each player is acting optimally. Equivalently again: No player can gain by deviating alone, i.e., by changing his or her strategy single-handedly.

In the game in Figure 11.5, the strategies (D, C) form a Nash equilibrium: if Player 1 plays D, Player 2 gets her best payoff by playing C; and if Player 2 plays C, Player 1 gets his best payoff by playing D. So the players' strategies are best responses to each other; equivalently, no player can gain by deviating alone. (Question: Are there any other Nash equilibria in this game?)

Algorithms for Finding Nash Equilibria

The best way to identify the Nash equilibria of a game is to first identify all of the outcomes that are *not* Nash equilibria; anything left must be a Nash equilibrium. For example, consider the game in Figure 11.5. The strategy pair (U, L) is not a Nash equilibrium because Player 2 can gain by deviating alone to R; (U, C) is not a NE because Player 1 can gain by deviating alone to D (and Player 2 can gain by deviating alone to L or R); etc. If you go through the options one by one and cross out those that are *not* Nash equilibria, the remaining options *will* be Nash equilibria (See Figure 11.6a).

A shortcut (but one you should use carefully!) is to underline each player's best responses.¹ To apply this to the game in Figure 11.5, first assume that Player 2 plays L; Player 1's best response is to play U, so underline the "5" in the box corresponding to (U, L). Next assume that Player 2 plays C; Player 1's best response is to play D, so underline the "3" in the box corresponding to (D, C). Finally, assume that Player 2 plays R; Player 1's best response is to play M, so underline the "4" in the box corresponding to (M, R). Now do the same thing for Player 2: go through all of Player 1's options and underline the best response for Player 2. (Note that C and R are both best responses when Player 1 plays M!) We end up with Figure 11.6b: the only boxes with both payoffs underlined are (D, C) and (M, R), the Nash equilibria of the game.

	L	C	R
U	5 / 1	2 / 0	2 / 2
M	0 / 4	1 / 5	4,5
D	2 / 4	3,6	1 / 0

(a)

	L	C	R
U	<u>5</u> ,1	2,0	2, <u>2</u>
M	0,4	1, <u>5</u>	<u>4</u> , <u>5</u>
D	2,4	<u>3</u> , <u>6</u>	1,0

(b)

Figure 11.6: Finding Nash equilibria: (a) with strike-outs; (b) with underlinings

¹It is easy to confuse the rows and columns and end up underlining the wrong things. Always double-check your answers by confirming that no player can gain by deviating alone.

(a)

		Player 2		
		L	C	R
Player 1	U	0,3	2,1	5,0
	M	4,8	3,2	8,3
	D	3,7	6,3	6,8

(b)

		Player 2		
		L	C	R
Player 1	U	-1,4	7,3	5,2
	M	2,0	5,-1	6,2
	D	1,2	1,0	1,0

(c)

		Player 2		
		L	C	R
Player 1	U	1,0	7,3	2,1
	M	1,0	1,2	6,2
	D	1,2	1,-3	1,0

(d)

		Player 2		
		L	C	R
Player 1	U	3,-1	5,4	3,2
	M	-2,5	1,3	2,1
	D	3,3	3,6	3,0

(e)

		Player 2		
		L	C	R
Player 1	U	3,-1	1,0	-1,-1
	M	1,-5	6,3	-7,-5
	D	-8,-10	-1,-3	-1,-1